# 1 Parabolic Equation Formulation

We consider the same setting as the wavenumber-integration formulation: an acoustic source at $\mathbf{r}_s = (0, 0, z_s)$ transmitting a CW signal at frequency $f$, with time dependence $e^{-i\omega t}$ and the depth axis pointing downwards from the sea surface at $z = 0$. The difference is that the sound speed is now allowed to vary with range as well as depth, $c = c(r, z)$. This is the situation the parabolic equation (PE) is built for.

Rotational symmetry about the source axis still holds, so the field obeys the cylindrical Helmholtz equation

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{p}}{\partial r}\right) + \frac{\partial^2 \bar{p}}{\partial z^2} + k^2(r,z)\,\bar{p} = -\frac{\delta(r)\,\delta(z-z_s)}{2\pi r}, \qquad k(r,z) = \frac{\omega}{c(r,z)}. \tag{1}$$

We fix a reference sound speed $c_0$, with reference wavenumber $k_0 = \omega/c_0$, and define the refractive index

$$n(r,z) = \frac{c_0}{c(r,z)} \quad \Longrightarrow \quad k^2(r,z) = k_0^2\, n^2(r,z). \tag{2}$$

## 1.1 Far-field factorisation

Away from the source the field is dominated by an outgoing cylindrical wave whose phase advances as $e^{ik_0 r}$. We factor this out explicitly and let a slowly varying envelope $\psi(r, z)$ carry everything else:

$$\bar{p}(r,z) = \psi(r,z)\, H_0^{(1)}(k_0 r). \tag{3}$$

Writing $H \equiv H_0^{(1)}(k_0 r)$ and primes for $d/dr$,

$$\begin{aligned} \frac{\partial \bar{p}}{\partial r} &= \psi_r H + \psi H', \\ \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{p}}{\partial r}\right) &= \psi_{rr} H + 2\psi_r H' + \psi H'' + \frac{1}{r}\left(\psi_r H + \psi H'\right). \end{aligned}$$

Since $H$ solves Bessel's equation, $H'' + H'/r + k_0^2 H = 0$, the terms in $\psi$ alone collapse to $-k_0^2 \psi H$:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{p}}{\partial r}\right) = \psi_{rr} H + \psi_r\left(2H' + \frac{H}{r}\right) - k_0^2 \psi H.$$

Substituting into (1) for $r > 0$, using (2), and dividing by $H$,

$$\psi_{rr} + \left(\frac{2H'}{H} + \frac{1}{r}\right)\psi_r + \psi_{zz} + k_0^2\left(n^2 - 1\right)\psi = 0. \tag{4}$$

This is still exact. In the far field, $k_0 r \gg 1$, the Hankel function takes its asymptotic form

$$H_0^{(1)}(k_0 r) \simeq \sqrt{\frac{2}{\pi k_0 r}}\, e^{i(k_0 r - \pi/4)} \quad \Longrightarrow \quad \frac{H'}{H} \simeq ik_0 - \frac{1}{2r},$$

so the coefficient of $\psi_r$ reduces to $2ik_0$ and

$$\psi_{rr} + 2ik_0\,\psi_r + \psi_{zz} + k_0^2\left(n^2 - 1\right)\psi = 0. \tag{5}$$

### 1.2 The paraxial approximation

Because the fast phase $e^{ik_0 r}$ has been removed, $\psi$ changes slowly over a wavelength in range. This is the paraxial approximation

$$\left|\frac{\partial^2 \psi}{\partial r^2}\right| \ll 2k_0 \left|\frac{\partial \psi}{\partial r}\right|, \tag{6}$$

valid when energy travels at small angles to the horizontal. Dropping $\psi_{rr}$ from (5) gives the standard parabolic equation (SPE) of Tappert,

$$2ik_0 \frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} + k_0^2 \left(n^2(r,z) - 1\right) \psi = 0, \tag{7}$$

or, solved for the range derivative,

$$\frac{\partial \psi}{\partial r} = \frac{i}{2k_0} \frac{\partial^2 \psi}{\partial z^2} + \frac{ik_0}{2} \left(n^2(r,z) - 1\right) \psi. \tag{8}$$

The name reflects what has happened mathematically. Equation (1) is elliptic: a boundary-value problem in which every point of the domain influences every other, including energy scattered back towards the source. Dropping $\psi_{rr}$ turns it into a parabolic equation: an initial-value problem in range, first order in $r$, in which the field at $r + \Delta r$ depends only on the field at $r$. Range plays the role of time, and (8) has the form of a Schrödinger equation with $\hbar = 1$, mass $k_0$ and potential $-\frac{k_0}{2}(n^2 - 1)$. The price is that backscattered energy is lost.

### 1.3 Operator factorisation and wide-angle equations

The SPE is accurate only to propagation angles of roughly 10° to 20° from the horizontal. A more systematic route, which also yields wide-angle equations, starts from the far-field form of (1). With $u = \sqrt{r}\,\bar{p}$,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial \bar{p}}{\partial r}\right) = \frac{1}{\sqrt{r}} \left(u_{rr} + \frac{u}{4r^2}\right),$$

and dropping $u/4r^2$ for $k_0 r \gg 1$ leaves, for $r > 0$,

$$\frac{\partial^2 u}{\partial r^2} + k_0^2\, Q^2 u = 0, \qquad Q = \sqrt{n^2 + \frac{1}{k_0^2}\frac{\partial^2}{\partial z^2}}. \tag{9}$$

If the environment varies slowly enough in range that $Q$ may be treated as independent of $r$ (its commutator with $\partial_r$ neglected), the operator factorises into outgoing and incoming parts,

$$\left(\frac{\partial}{\partial r} + ik_0 Q\right)\left(\frac{\partial}{\partial r} - ik_0 Q\right) u = 0. \tag{10}$$

Keeping only the outgoing factor gives the one-way equation $\partial_r u = ik_0 Q u$. In the far field $u \propto \psi\, e^{ik_0 r}$ by (3), so

$$\frac{\partial \psi}{\partial r} = ik_0 \left(Q - 1\right) \psi, \qquad Q = \sqrt{1 + q}, \qquad q = n^2 - 1 + \frac{1}{k_0^2}\frac{\partial^2}{\partial z^2}. \tag{11}$$

Equation (11) is exact for a range-independent medium, apart from the far-field step. Each PE corresponds to a choice of rational approximation to the square-root operator:

| Approximation | $Q = \sqrt{1+q} \approx$ | Angular validity |
|---|---|---|
| Tappert (SPE) | $1 + \frac{1}{2}q$ | $\sim 10°$–$20°$ |
| Claerbout | $\dfrac{1+\frac{3}{4}q}{1+\frac{1}{4}q}$ | $\sim 40°$ |
| Padé, $m$ terms (Collins) | $1 + \displaystyle\sum_{j=1}^{m} \frac{a_j\, q}{1 + b_j\, q}$ | up to $\sim 90°$ |

with Padé coefficients

$$a_j = \frac{2}{2m+1} \sin^2\!\left(\frac{j\pi}{2m+1}\right), \qquad b_j = \cos^2\!\left(\frac{j\pi}{2m+1}\right). \tag{12}$$

Substituting the Tappert form into (11) recovers the SPE (8) exactly, which confirms that both routes agree.

### 1.4 Boundary conditions, seabed and starting field

The pressure-release surface requires

$$\psi(r, 0) = 0. \tag{13}$$

Unlike the rigid bottom of the wavenumber-integration formulation, the seabed is now a penetrable fluid half-space with sound speed $c_b$, density $\rho_b$ and attenuation $\alpha_b$ (in dB per wavelength). Energy reaching it at steep angles is partly transmitted, so the computational domain is continued below the seabed at $z = D$ into the sediment, down to $z_{\max}$. Since the domain must end somewhere, an artificial absorbing layer (a sponge) occupies $z_{\mathrm{sp}} < z < z_{\max}$, in which the attenuation is ramped up smoothly so that downgoing energy is absorbed rather than reflected back:

$$k(z) = \frac{\omega}{c(z)} \left(1 + \frac{i\,\alpha(z)}{40\pi \log_{10} e}\right), \qquad \alpha(z) = \alpha_b + \alpha_{\max} \left(\frac{z - z_{\mathrm{sp}}}{z_{\max} - z_{\mathrm{sp}}}\right)^2 \quad (z > z_{\mathrm{sp}}), \tag{14}$$

with $\psi(r, z_{\max}) = 0$ at the base. At the seabed itself, pressure and normal particle velocity are continuous:

$$\psi \text{ continuous}, \qquad \frac{1}{\rho}\frac{\partial \psi}{\partial z} \text{ continuous at } z = D. \tag{15}$$

The density jump is what makes the depth operator $\rho\, \partial_z\big(\rho^{-1} \partial_z\big)$ rather than $\partial_z^2$, and it is the reason for the choice of solver in the next section.

Because the PE is first order in range it needs a starting field $\psi(0, z)$ in place of the point source. Greene's source covers a wider angular spectrum than a Gaussian beam and is used here, with an image term enforcing (13):

$$\psi(0, z) = \sqrt{k_0} \left[ \big(1.4467 - 0.4201\, k_0^2 (z - z_s)^2\big)\, e^{-\frac{k_0^2 (z - z_s)^2}{3.0512}} - \big(1.4467 - 0.4201\, k_0^2 (z + z_s)^2\big)\, e^{-\frac{k_0^2 (z + z_s)^2}{3.0512}} \right]. \tag{16}$$

### 1.5 Finite-difference Padé solution

**Why not split-step Fourier.** The split-step Fourier method splits the SPE (8) into a diffraction step, diagonal in vertical wavenumber, and a refraction step, diagonal in depth, each costing one

FFT. In a uniform medium it is essentially exact: against the image-source solution for a pressure-release surface over a half-space of water, it agrees to 0.01 dB. But a Fourier method treats the depth operator as $\partial_z^2$ on a periodic domain, so it cannot represent the interface conditions (15). Smoothing the density step and folding it into an effective index does not rescue it: for the waveguide studied here ($D = 200\,\mathrm{m}$, $f = 100\,\mathrm{Hz}$, $c_b = 1650\,\mathrm{m/s}$, $\rho_b/\rho_w = 1.8$) the split-step Fourier field is 2–6 dB rms in error from 1 to 4 km. The seabed is therefore handled by a finite-difference discretisation instead.

**Depth discretisation.** The domain is sampled at nodes $z_i = i\,\Delta z$, $i = 1, \ldots, N$, with $\psi_0 = \psi_{N+1} = 0$ and the seabed placed exactly on node $i_D$ ($D = i_D \Delta z$). The operator $X$ of (11), generalised to variable density, is discretised as

$$(X\psi)_i = \frac{1}{k_0^2 \Delta z^2} \left[ \frac{\rho_i}{\rho_{i+\frac{1}{2}}} (\psi_{i+1} - \psi_i) - \frac{\rho_i}{\rho_{i-\frac{1}{2}}} (\psi_i - \psi_{i-1}) \right] + \frac{k_i^2 - k_0^2}{k_0^2} \psi_i, \tag{17}$$

with the density evaluated at the half nodes $z_{i\pm\frac{1}{2}}$. Because the interface falls between half nodes, the jump in $\rho^{-1}\partial_z\psi$ across it is represented exactly, which is precisely what the Fourier method could not do. On the interface node the wavenumber is averaged, $k_{i_D}^2 = \frac{1}{2}\left(k_w^2 + k_b^2\right)$. The result is a tridiagonal matrix $X$.

**Range marching.** The square root in (11) is replaced by the $m$-term Padé series (12), and the exponential of the sum is split into a product of its terms, applied in sequence:

$$\psi(r + \Delta r) = e^{\,ik_0 \Delta r\,(\sqrt{1+X}-1)}\,\psi(r) \approx \prod_{j=1}^{m} \exp\!\left( \frac{ik_0 \Delta r\, a_j X}{1 + b_j X} \right) \psi(r). \tag{18}$$

Each factor is advanced with the Crank–Nicolson (Cayley) approximation $e^x \approx (1+x/2)/(1-x/2)$. Multiplying through by $1 + b_j X$ gives, with $\sigma = k_0 \Delta r$,

$$\underbrace{\left[1 + \left(b_j - \tfrac{i\sigma a_j}{2}\right) X\right]}_{A_j} \psi' = \underbrace{\left[1 + \left(b_j + \tfrac{i\sigma a_j}{2}\right) X\right]}_{B_j} \psi. \tag{19}$$

Each sub-step is one tridiagonal multiplication and one tridiagonal solve, and since $A_j$ does not change from step to step its LU factors are computed once. For real $X$, $A_j$ and $B_j$ are complex conjugates, so each sub-step has unit gain on every eigenvector of $X$: the scheme is unconditionally stable, and evanescent components are carried without growth. The field is then recovered from (3),

$$\bar{p}(r, z) = s\,\psi(r, z)\,H_0^{(1)}(k_0 r), \tag{20}$$

where $s \in \mathbb{C}$ absorbs the source level and phase.

**Accuracy.** Table 1 compares the march with a converged wavenumber integral for the same environment, as an RMS transmission-loss difference over 1–4 km at three receiver depths. The Padé order matters (the single-term Claerbout equation is 1–3 dB off), the range step matters little, and the depth step dominates what remains; $m = 4$, $\Delta z = 0.5\,\mathrm{m}$ and $\Delta r = 2\,\mathrm{m}$ is used as the default.

| Settings | TL error (dB rms) |
|---|---|
| Split-step Fourier, $\Delta z = 0.5\,\mathrm{m}$ | 2–6 |
| Padé $m = 1$ (Claerbout), $\Delta z = 0.5\,\mathrm{m}$, $\Delta r = 2\,\mathrm{m}$ | 0.7–3.0 |
| Padé $m = 4$, $\Delta z = 1\,\mathrm{m}$, $\Delta r = 2\,\mathrm{m}$ | 0.6–1.5 |
| Padé $m = 4$, $\Delta z = 0.5\,\mathrm{m}$, $\Delta r = 4\,\mathrm{m}$ | 0.3–0.7 |
| Padé $m = 4$, $\Delta z = 0.5\,\mathrm{m}$, $\Delta r = 2\,\mathrm{m}$ | 0.3–0.5 |
| Padé $m = 4$, $\Delta z = 0.25\,\mathrm{m}$, $\Delta r = 2\,\mathrm{m}$ | 0.1–0.3 |

Table 1: Error of the PE march against a converged wavenumber integral ($D = 200\,\mathrm{m}$, $f = 100\,\mathrm{Hz}$, $c_b = 1650\,\mathrm{m/s}$, $\rho_b/\rho_w = 1.8$, $z_s = 40\,\mathrm{m}$), over 1–4 km at depths of 30, 100 and 170 m.

### 1.6 Data-aided parabolic-equation formulation

As in the data-aided wavenumber-integration formulation, the environment is taken as unknown and sparse measurements $y_j$ are available at locations $(r_j, z_j)$. In the wavenumber formulation the trainable parameters were the spectral kernel values $g(k_{r,m}, z_l)$ themselves; in the PE the propagation physics is carried by the march (19), so what remains unknown is the environment that enters $X$. The water sound speed is represented by a bounded one-hidden-layer network,

$$c(z;\boldsymbol{\theta}) = c_{\min} + (c_{\max} - c_{\min})\,\mathrm{sigmoid}\Big(\mathbf{W}_2^{\top}\,\mathrm{ReLU}(\mathbf{W}_1\tilde{z} + \mathbf{b}_1) + b_2\Big), \qquad \tilde{z} = \frac{2z}{D} - 1, \tag{21}$$

and the seabed parameters are bounded in the same way, so the set of trainable parameters is

$$\mathcal{T} \equiv \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, b_2,\ c_b,\ \rho_b,\ \alpha_b\}, \tag{22}$$

about thirty numbers in total. The source factor $s$ enters the field linearly, so it is not trained but solved exactly at every evaluation by least squares,

$$s^{\star} = \frac{\mathbf{p}^{\mathsf{H}}\mathbf{y}}{\mathbf{p}^{\mathsf{H}}\mathbf{p}}, \qquad p_j = \psi(r_j, z_j)\,H_0^{(1)}(k_0 r_j), \tag{23}$$

and the parameters are learned by minimising the relative complex-field misfit

$$\mathcal{L}(\mathcal{T}) = \frac{\sum_j |s^{\star}p_j - y_j|^2}{\sum_j |y_j|^2}. \tag{24}$$

The model assumes $e^{-i\omega t}$ time dependence, for which $H_0^{(1)}$ is the outgoing wave; data in the opposite convention are the complex conjugate, and are conjugated before training. The misfit also has local minima in $c_b$, which controls the critical angle and hence which energy remains trapped, so training starts from the best point of a coarse scan over $c_b$.

### 1.7 Adjoint gradient

Differentiating the march with forward-mode automatic differentiation costs one march per parameter. Instead, the gradient is obtained from the adjoint of the march, at the cost of one forward march and one backward march regardless of the number of parameters.

Write the march as a sequence of sub-steps $\psi_{n+1} = A^{-1}B\,\psi_n$, one per Padé term per range step, with $A = 1 + c_l X$ and $B = 1 + c_r X$ as in (19). A perturbation $\delta X$ of the operator changes a sub-step by

$$\delta\psi_{n+1} = A^{-1}\left(B\,\delta\psi_n + c_r\,\delta X\,\psi_n - c_l\,\delta X\,\psi_{n+1}\right) = A^{-1}\left(B\,\delta\psi_n + \delta X\,\mathbf{v}_n\right), \qquad \mathbf{v}_n = c_r\psi_n - c_l\psi_{n+1}. \tag{25}$$

For a real loss, write $\delta\mathcal{L} = 2\,\mathrm{Re}\left(\boldsymbol{\lambda}_{n+1}^{\mathsf{H}}\,\delta\psi_{n+1}\right)$, where $\boldsymbol{\lambda}_{n+1} = \partial\mathcal{L}/\partial\psi_{n+1}^{*}$ is the sensitivity to the field after the sub-step. Introducing $\boldsymbol{\mu}_n = A^{-\mathsf{H}}\boldsymbol{\lambda}_{n+1}$ and substituting (25),

$$\delta\mathcal{L} = 2\,\mathrm{Re}\left(\boldsymbol{\mu}_n^{\mathsf{H}} B\,\delta\psi_n\right) + 2\,\mathrm{Re}\left(\boldsymbol{\mu}_n^{\mathsf{H}}\,\delta X\,\mathbf{v}_n\right). \tag{26}$$

The first term gives the sensitivity before the sub-step, and the second the sensitivity to the operator. The adjoint march therefore runs backwards from the last sub-step:

$$\boldsymbol{\mu}_n = A^{-\mathsf{H}}\boldsymbol{\lambda}_{n+1}, \qquad \boldsymbol{\lambda}_n = B^{\mathsf{H}}\boldsymbol{\mu}_n, \qquad G_{i,i'} \mathrel{+}= \mu_{n,i}\,v_{n,i'}^{*} \quad (|i-i'|\le 1), \tag{27}$$

where $G_{i,i'}$ accumulates the sensitivity to the tridiagonal entry $X_{i,i'}$, so that $\delta\mathcal{L} = 2\,\mathrm{Re}\sum G_{i,i'}^{*}\,\delta X_{i,i'}$. Each backward sub-step costs one solve with $A^{\mathsf{H}}$, whose LU factors are again computed once, and one multiplication by $B^{\mathsf{H}}$. The march is seeded at the range steps that the receivers interpolate from: since $p_j$ is linear in $\psi$, the sensitivity

$$\frac{\partial\mathcal{L}}{\partial p_j^{*}} = \frac{s^{\star\star}\,(s^{\star}p_j - y_j)}{\sum_k |y_k|^2} \tag{28}$$

is multiplied by $H_0^{(1)}(k_0 r_j)^{*}$ and spread back onto the neighbouring depth nodes and range steps with the same interpolation weights. Because $s^{\star}$ minimises the misfit, its own dependence on the parameters does not contribute to the gradient.

Finally, the entries of $X$ depend on the parameters only through (17), (14) and (21), which involve no march, so the last link

$$\frac{\partial\mathcal{L}}{\partial\theta} = 2\,\mathrm{Re}\sum_{|i-i'|\le 1} G_{i,i'}^{*}\,\frac{\partial X_{i,i'}}{\partial\theta} \tag{29}$$

is cheap to evaluate by any means. The adjoint gradient agrees with a central finite difference to about $10^{-7}$ relative error.

Compared with wavenumber integration, the PE gives up backscatter, but its march applies the environment one range step at a time, so a range-dependent environment only requires $X$ to be rebuilt along the way, at no change to the method or the adjoint.